1. Clinical Question & Core Hypothesis

Clinical Question

EMT/Wnt-driven programs promote metastasis yet some tumors with abundant lymphocytes fail to respond to immunotherapy — why does immune infiltration not translate into therapeutic benefit? 2) Built on the single-path mechanistic chain provided: EMT → Wnt/beta-catenin signaling → modulation of CXCR4 → upregulation of GLS → regulation of Cyclin D1; this links migration/EMT, tumor glutamine metabolism (GLS), and proliferative drive (Cyclin D1) to possible immunotherapy resistance. 3) Supporting relationships: EMT can alter chemokine receptor signaling (CXCR4) which correlates with GLS; GLS activity influences Cyclin D1; aggressive, protein-overexpressing tumors (here exemplified by CCND1-driven MCL) have been reported to derive less benefit from immunotherapy, consistent with a metabolism-mediated resistance mechanism. 4) Endpoint choice: therapy_response (immunotherapy) was chosen because the path explicitly implicates reduced benefit from immunotherapy and because the mechanistic question asks under what metabolic/proliferative conditions immune infiltration fails. 5) Deliberate exclusions: I excluded chemotherapy/drug details (docetaxel, nedaplatin), transporter relationships, and broader pathway nodes (unspecified WNT ligands or beta-catenin gene names) to keep the hypothesis sharply focused on the metabolic–proliferative mechanism that could decouple TIL presence from effective immunotherapy response?

Working Hypothesis

Does breast cancer with high imaging-derived TIL density exhibit a reduced predicted immunotherapy response in tumors that are GLS-high and CCND1-high, indicating a metabolic–proliferative state that decouples immune infiltration from therapeutic benefit? Is CXCR4 signaling a mechanistic mediator of this decoupling, such that CXCR4 activity strengthens the negative association between GLS/CCND1 status and predicted immunotherapy response in TCGA-BRCA samples?

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	GLS, CCND1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

GLS: GLS; GLS activity influences Cyclin D1; aggressive, protein-overexpressing tumors (here exemplified by CCND1-driven MCL) have been reported to derive less benefit from immunotherapy, consistent with a metabolism-mediated resistance mechanism.

CCND1: CCND1-driven MCL) have been reported to derive less benefit from immunotherapy, consistent with a metabolism-mediated resistance mechanism.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	CXCR4	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: cohort stratification by GLS and CCND1 expression (high/low) and CXCR4 status;

Prediction 2: assessment of imaging-derived TIL density;

Prediction 3: modeling of predicted immunotherapy response using transcriptomic immune signatures;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CXCR4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.